

Optimizing Input Data for Gridding Climate Normals for Canada

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ABSTRACT

Spatial models of 1971–2000 monthly climate normals for daily maximum and minimum temperature and total precipitation are required for many applications. The World Meteorological Organization's recommended standard for the calculation of a normal value is a complete 30-yr record with a minimal amount of missing data. Only 650 stations (~16%) in Canada meet this criterion for the period 1971–2000. Thin-plate smoothing-spline analyses, as implemented by the Australian National University Splines (ANUSPLIN) package, are used to assess the utility of differing amounts of station data in estimating nationwide monthly climate normals. The data include 1) only those stations (1169) with 20 or more years of data, 2) all stations (3835) with 5 or more years of data in at least one month, and 3) as in case 2 but with data adjusted through the most statistically significant linear-regression relationship with a nearby long-term station to 20 or more years (3983 stations). Withheld-station tests indicate that the regression-adjusted normals as in dataset 3 generally yield the best results for all three climatological elements, but the unadjusted normals as in dataset 2 are competitive with the adjusted normals in spring and autumn, reflecting the known longer spatial correlation scales in these seasons. The summary mean absolute differences between the ANUSPLIN estimates and the observations at 48 spatially representative withheld stations for dataset 3 are 0.36°C, 0.66°C, and 4.7 mm, respectively, for maximum temperature, minimum temperature, and precipitation. These are respectively 18%, 7%, and 18% smaller than the summary mean absolute differences for the long-term normals in dataset 1.

1. Introduction

Spatial models of climate normals are required for a wide range of applications. The World Meteorological Organization (WMO) defines a climate normal as an

arithmetic mean for a fixed 30-yr period ending in a “tens” year (WMO 1989). For a monthly datum to be included in a temperature-normal calculation, no more than three consecutive days and no more than five days in total can be missing for the month. For total precipitation, daily data must be complete for the whole month to be acceptable. A full type-A normal (Table 1) applies the same principle that no more than three consecutive years nor any five years can be missing for the calculation of a monthly normal. Most station data in Canada

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TABLE 1. Types of monthly normals calculated by Environment Canada for 1971–2000.

Normal code	No. of years with valid monthly data required in the 1971–2000 period
A	WMO “3 and 5 rule” (i.e., no more than 3 consecutive years and no more than 5 total missing years for either temperature or precipitation).
B	At least 25 years of monthly values
C	At least 20 years of monthly values
D	At least 15 years of monthly values
E	At least 10 years of monthly values
F	At least 5 years of monthly values
G	Fewer than 5 years of monthly values

fail to meet this criterion. The use of normals calculated with fewer years of data presents the risk that the resulting values may not be representative of the full 30-yr period.

In the development of spatial models, decisions are required concerning greater station density versus more complete records. In Canada, as elsewhere, there are choices to be made when deciding on the station data to be used for such models. Here we examine this issue by assessing predictive errors of spatial models derived from different candidate datasets. The spatial models are derived using thin-plate smoothing splines as implemented by the Australian National University Splines (ANUSPLIN) package (Hutchinson 1991, 1995, 2011). ANUSPLIN has been widely used by researchers around the world, including applications in Canada for about two decades. Spatial extension of climate normals can be done by a range of objective analysis techniques, from inverse-distance weighting (Cressman 1959), to kriging and smoothing splines, to quasi-statistical techniques such as the Parameter-Elevation Regressions on Independent Slopes Model (PRISM; Daly et al. 1994, 1997, 2002). Previous studies by the authors have led to the development of a wide range of spatial models covering various spatial and temporal extents (e.g., McKenney et al. 2006, 2011; Price et al. 2000; Hutchinson et al. 2009). Milewska et al. (2005) evaluated a number of spatial-modeling approaches and found that ANUSPLIN and PRISM yielded similar results for gridded 1961–90 normal maximum and minimum temperature and precipitation over the Canadian prairies. Although all of these efforts have involved extensive testing, none have documented the relationship between completeness of record for monthly normals and model quality—a general rule of thumb (Environment Canada 1986a,b; Natural Resources Canada 2011a,b) has been that more station data are preferred even if the period of record is incomplete, particularly in Canada where station density is low in many regions (Fig. 1).

Given the spatial and temporal limitations of the Canadian climate network, it is important to optimize the use of short- and long-term climate normals in the generation of spatial grids of monthly maximum temperature, minimum temperature, and precipitation normals for the country. In this paper the data choices are described and a brief description of the ANUSPLIN spatial analysis tool is also provided. Spatial analyses derived from most data-fitting techniques, including the ANUSPLIN package, will agree well in general with the values at the observation points in the supplied dataset. Predictive skill at locations other than the input data points can be tested by withholding an appropriate subset of data from the model building phase. We explain our approach to withholding stations and utilize both standard model diagnostics and the predictive success of estimates at the withheld station locations to assess the impact of varying the number and data completeness of stations. The results of the diagnostic and withheld data tests are presented and discussed for each climate element.

2. Data and methods

When Environment Canada (2011) calculates the temperature and precipitation normals for the climate observing stations in Canada for 1971–2000, it classifies various types of monthly normals as shown in Table 1. For Canada, there are over 3000 stations with type-F normals or better (5 or more years) for precipitation and over 2000 stations for temperature during the period 1971–2000. This translates to $\sim 2857 \text{ km}^2$ per station for precipitation and $\sim 4755 \text{ km}^2$ per station for temperature as compared with $\sim 1076 \text{ km}^2$ per station for precipitation and $\sim 1520 \text{ km}^2$ per station for temperature in the contiguous United States (Janis et al. 2004). In the ideal situation, these stations would be distributed evenly across Canada, but most are located near the Canada–U.S. border (Fig. 1). In fact, as noted by Hutchinson et al. (2009), 95% of the stations are situated in the southern half of the country. Alberta has the best spatial coverage of any province or territory in Canada, but many of its stations in the north and in the foothills operate in summer only. In British Columbia there are very few high-elevation climate stations, and most stations are located in valley bottoms. The station network is also very sparse over northern Canada north of 60°N and east of the Mackenzie River valley, over northern British Columbia, and from northern Saskatchewan to northern Quebec. If only type-A normals were considered then there would be just 650 highly asymmetrically distributed stations to represent the climate of the country, but there would be high confidence that these values would

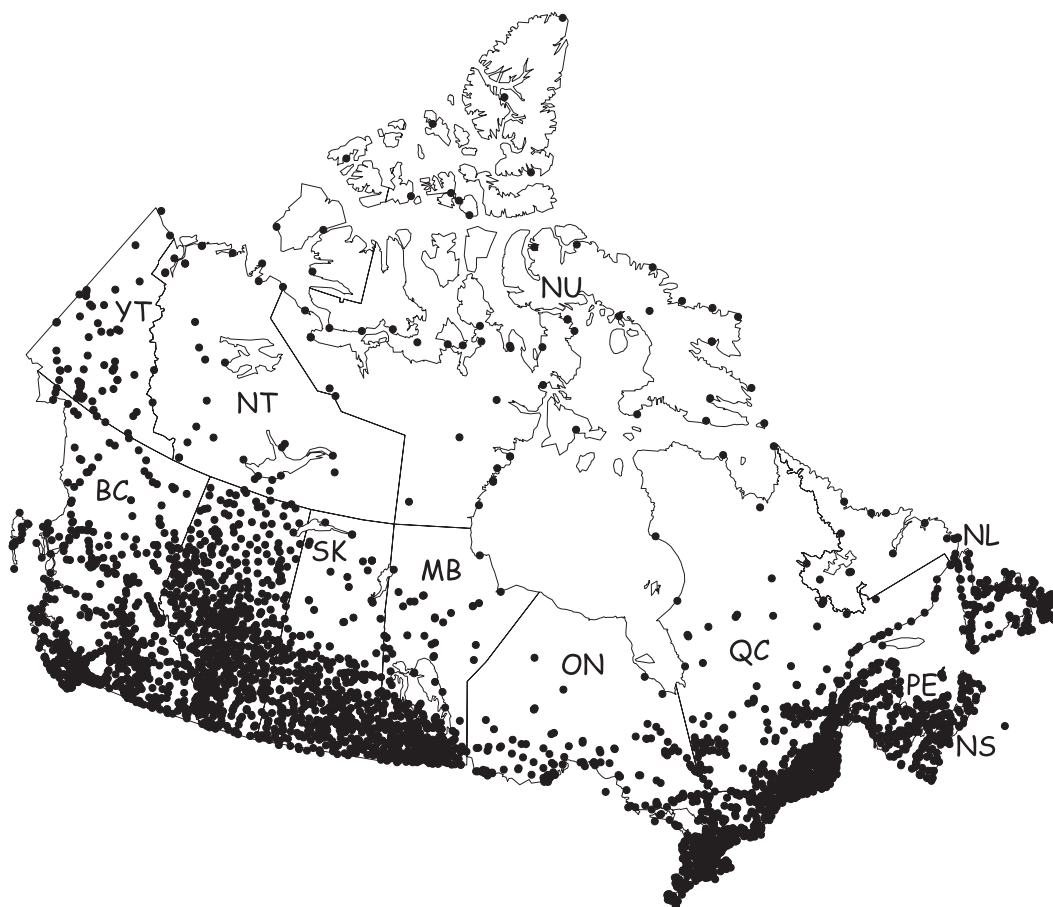


FIG. 1. Distribution of 1971–2000 monthly-normal precipitation stations.

provide the best estimates of the normals at these specific locations. If all stations with 5 or more years were utilized for precipitation, then over 3000 stations would be available but the temporal representativeness of the short-period stations would be uncertain.

All normal types were used in this study except type-G normals (fewer than 5 years of data), which were eliminated from further consideration as having records that are much too short to provide anything close to a reliable estimate of the normal. In general, the danger of accepting normals calculated with fewer years is that the short-period average may differ substantially from the population mean because of trends in the underlying data or anomalous periods sampled with short-period records. For example, there has been a statistically significant trend in temperature over western Canada in the winter and spring months during the period 1971–2000 (Zhang et al. 2000, 2010). A short-period sample from the start or the end of the period will result in a biased normal being calculated. Both short-period temperature and precipitation normals can be affected by anomalous periods, especially normals calculated

from as few as 5 years of data. Historic droughts and wet spells can persist for several years (Roberts et al. 2006; Sauchyn et al. 2002), and paleoclimatic studies (Sauchyn 2003; St. George et al. 2009) suggest that decade-long droughts have occurred in the past in Canada.

Other potential influences on climate normals include unrepresentative local effects (frost hollows), systematic observer bias and errors, archive errors, climatological day bias (Vincent et al. 2009; Hopkinson et al. 2011) for minimum and maximum temperature at principal climate stations, and instrument calibration and exposure. Elevation also exerts a strong influence on both temperature and precipitation, but this effect is accounted for in our ANUSPLIN analyses, which include elevation as an independent variable.

All spatial models of monthly climate normals in this study have been fitted using thin-plate smoothing splines as implemented by ANUSPLIN (Hutchinson 2011). The mathematical theory behind thin-plate smoothing splines can be found in Wahba (1990), and other studies contain further details on its application to climate data in numerous applications by Hutchinson and

colleagues (e.g., Hutchinson and Bischof 1983; Hutchinson 1991, 1995, 1998, 2011; Hutchinson et al. 2009; Hutchinson and Gessler 1994); see also McKenney et al. (2001, 2006, 2011) and Price et al. (2000) for Canadian applications. Central features of ANUSPLIN are its ability to incorporate spatially varying dependencies on elevation and its application of data smoothing to minimize predictive error for any given location. The residuals of the fitted ANUSPLIN surfaces from the data reflect both data errors and the degree of spatial representativeness of the data. Standard output from ANUSPLIN provides several measures of model quality. Of key interest here are the signal, the generalized cross validation (GCV), and residuals from withheld data.

The signal, provided by the trace of the influence matrix, indicates the complexity of the surface and varies between a small positive integer and the number of stations used to generate the model (Wahba 1990; Hutchinson 2011). In general, the signal should be no greater than one-half of the number of data points (Hutchinson and Gessler 1994). Models with appropriate signals indicate a stable dependence on the independent variables (in this case longitude, latitude, and suitably scaled elevation). Such models tend to be more reliable (accurate) in data-sparse regions.

The GCV is a measure of predictive error of the fitted surface that is calculated by (implicitly) removing each data point in turn and summing, with appropriate weighting, the square of the difference of each omitted data value from the spline fitted to all of the remaining data points. It plays two roles here. The first is to optimize the degree of data smoothing in the fitted model. The second is to provide a direct measure of predictive error. The square root generalized cross validation (RtGCV) is usually a good indicator of the standard predictive error of the fitted surface. As discussed by Hutchinson (1998) and by McKenney et al. (2011), the GCV can perform less well in the presence of data with significant short-range correlation and can be somewhat biased when applied to unevenly spaced data networks dominated by particular data-dense areas. The latter can be overcome by explicitly withholding a spatially representative subset of station data from the model-building procedure and comparing model estimates of the observed values at these locations. This provides an unbiased estimate of the predictive accuracy of the models. The withheld locations were obtained here by using the “SELNOT” (from “knot selection”) procedure in the ANUSPLIN package to equisample the longitude, latitude, and suitably scaled elevation space of the long-term reference climate stations. This approach ensured that the withheld data were representative of the latitude, longitude, and elevation gradients across the southern half of the country (see Hutchinson

et al. 2009). The withheld stations were the same as those described by Hutchinson et al. (2009) except for the removal of two west-coast very-high-rainfall stations whose precipitation statistics would have dominated the summary statistics. The ensuing withheld standard predictive errors normally display good agreement with RtGCV values for countrywide analyses (McKenney et al. 2006; Hutchinson et al. 2009), especially for temperature, which is more reliably interpolated from limited data networks. We calculate both mean errors and mean absolute errors for the 48 spatially representative long-term stations that are shown in Fig. 2.

Three datasets—D20, D05, and D05adj—are examined here:

- 1) D20 refers to stations with type-C (20 yr or greater) normals (1169 precipitation stations).
- 2) D05 refers to stations with type-F normals in at least one month but unadjusted for sample length and comprises 3835 precipitation stations.
- 3) D05adj refers to the same stations as described for D05 but with short-period normals extended by regression with a nearby long-period station. All stations with fewer than the full 30 yr of data were separately regressed against each of the nearest five long-term stations within 1000 km by using simple linear regression. The large radius of influence was required for stations in northern Canada. For precipitation, the data were transformed to square root space for the regression analysis. If the short-term stations had data for fewer than five years, they were eliminated from further consideration. For each long-term station, all monthly pairs were used in the regression analysis rather than developing individual regressions for each month. The sample size had to be at least five. Using the *F* statistic, only those regressions that were significant at the 1% level of risk were accepted, and, of these, the regression with the greatest *F* value at one of the five long-term stations was chosen to estimate values for missing months at the short-term station. This procedure was applied only to those months for which there were at least four observed monthly values in the 30-yr-normals period. The adjusted normals were then calculated by utilizing the observed and estimated data. Often the adjusted normals were based on the full 30 years from 1971 to 2000, and in most cases they were based on well in excess of 20 years. Adjusted normals that were based on fewer than 30 years only occurred when the best nearby long-term station did not have a complete 30-yr record. This resulted in regression-adjusted normals for 3983 precipitation stations. This approach differs from that used by Sun



FIG. 2. Withheld stations for temperature and precipitation.

and Peterson (2005, 2006) to calculate pseudonormals for short-term stations in the 1981–2010 climate normals for the United States. Especially in northern Canada, there were too few stations to attempt their method and too few stations with full 30-yr normals.

The numbers of temperature stations in the three datasets D20, D05, and D05adj were proportionately less than the numbers of precipitation stations.

For the purposes of direct comparison, the same thin-plate smoothing-spline analysis, as provided by the “SPLINB” program in the ANUSPLIN package [SPLINB is a program that fits an arbitrary number of (partial) thin-plate smoothing-spline functions of one or more independent variables and is an approximation that is designed for larger data sets], was applied to each dataset. In the case of precipitation, the data were first transformed within SPLINB by applying the square root transformation. As discussed by Hutchinson et al. (2009), this removes the natural skew in precipitation data and balances the spatial variability between small and large precipitation values over the whole data network.

ANUSPLIN automatically removes the small negative bias in the back-transformed interpolated values that this transformation introduces (Hutchinson 2011). It also automatically transforms the output error statistics back to the original units of the data, as reported in Table 2. The SPLINB procedure uses all of the available data but limits the complexity of the fitted spline to that defined by a set of knots. These were selected from the data points by the SELNOT program to equisample the longitude, latitude, and suitably scaled elevation space spanned by the station network. As described by Hutchinson (1995) and Hutchinson et al. (2009), the elevations are scaled by a factor of 100 relative to the scaling of the horizontal coordinates, in keeping with the relative scales of horizontal and vertical atmospheric dynamics (Daley 1991). After initial assessments, SELNOT was used to select 50% of each dataset as knots. This threshold permitted models of reasonable complexity for each analysis but limited the signal to no more than one-half of the number of data points, ensuring the stable behavior described above. Withheld tests were run for each of the three datasets.

TABLE 2. Monthly model diagnostics summarized by season and year for full-data runs for monthly maximum temperature, minimum temperature, and precipitation normals for 1971–2000 for Canada (highest values are in italics, and lowest values are in boldface).

Season	RtGCV			RtMSR			Signal		
	D20	D05	D05adj	D20	D05	D05adj	D20	D05	D05adj
Max temperature (°C)									
Winter	0.57	<i>0.97</i>	0.52	0.37	<i>0.74</i>	0.35	383	681	1027
Spring	0.58	<i>0.79</i>	0.57	0.38	<i>0.61</i>	0.39	400	694	982
Summer	0.71	<i>0.78</i>	0.68	0.45	<i>0.55</i>	0.47	457	906	1018
Autumn	0.45	<i>0.64</i>	0.49	0.29	<i>0.49</i>	0.34	397	715	979
All	0.59	<i>0.80</i>	0.57	0.38	<i>0.60</i>	0.39	409	749	1002
Min temperature (°C)									
Winter	1.13	<i>1.37</i>	1.02	0.81	<i>1.02</i>	0.71	312	729	942
Spring	0.82	<i>0.91</i>	0.83	0.62	<i>0.69</i>	0.59	274	698	899
Summer	<i>0.90</i>	<i>0.90</i>	0.87	<i>0.68</i>	<i>0.68</i>	0.64	311	769	859
Autumn	0.92	<i>0.96</i>	0.89	0.70	<i>0.71</i>	0.64	268	756	878
All	0.95	<i>1.05</i>	0.90	0.71	<i>0.79</i>	0.65	291	738	895
Precipitation (mm)									
Winter	15.1	<i>15.4</i>	12.9	10.0	<i>10.9</i>	8.8	447	953	1133
Spring	10.3	<i>11.8</i>	9.3	6.7	<i>8.4</i>	6.3	485	983	1197
Summer	8.3	<i>11.6</i>	8.4	5.4	<i>8.8</i>	5.8	530	857	1192
Autumn	13.1	<i>14.4</i>	11.3	8.5	<i>9.9</i>	7.5	488	1035	1251
All	12.0	<i>13.4</i>	10.6	7.9	<i>9.6</i>	7.2	488	957	1193

3. Results

a. Full-dataset runs

Table 2 summarizes by season and year the monthly model diagnostics for applying SPLINB as explained above to spatially interpolate the full monthly datasets with no data withheld. There are systematic differences in the reported statistics for each dataset in keeping with their definitions. The D20 data have mostly higher signals relative to the number of data points, especially for maximum temperature and precipitation. This is to be expected since the D20 data are generally the most accurate and moreover are around one-third of the size of the other datasets. Thus each data point has to support interpolation over a larger surrounding area. The generally smaller signals, in absolute terms, for the D20 data indicate that the surfaces fitted to these data are relatively smooth and less able to represent finer-scale climatic gradients. Conversely, the higher signals in absolute terms for the D05 and D05adj data suggest that more information has been extracted from both of these datasets—in particular, for the D05adj data.

The root-mean-square residuals from the data (RtMSRs) in Table 2 are the residuals of the smoothed data values calculated by SPLINB from the data values. They reflect the errors in the data values in relation to the 1971–2000 normals. The RtMSRs for the D20 and D05adj datasets are in broad agreement, particularly for maximum temperature, and are systematically smaller than the RtMSRs for the D05 dataset. This suggests that the

regressions have generally reduced the errors in the short-period means to about the same level as the errors in the longer-period means and perhaps to an even-smaller level for the minimum temperature and precipitation data. The larger RtMSRs for the D05 datasets also confirm that these contain the noisiest data when considered as estimates of the 30-yr normals.

The RtGCVs assess surface accuracy against the data provided in each dataset. These assessments are therefore dependent on both the accuracy of the data and the accuracy of the fitted surfaces. Thus D05, the noisiest dataset, also has the largest RtGCVs for every season. In light of the similar errors of the D20 and D05adj datasets, the generally smaller RtGCVs for the D05adj dataset now indicate that this dataset has given rise to more accurate interpolated normals than has the smaller D20 dataset. The possible exceptions are for maximum temperatures in the autumn and precipitation in summer, for which the RtGCVs for the D20 data are systematically smaller than those for the D05adj data, but this result is likely due to the fact that the D20 RtMSRs are distinctly smaller than the D05adj RtMSRs in these cases.

Less clear is whether the predictive errors of the surfaces fitted to the D05 data are smaller than the predictive errors of the surfaces fitted to the D05adj data, since the surfaces that are based on the D05 data are being assessed with GCV against noisier data. The generally higher signals for the D05adj data suggest that more information has been extracted from these data.

TABLE 3. The monthly ME, MAE, and RMSE error statistics summarized by season and year for withheld-station tests for monthly maximum temperature, minimum temperature, and precipitation normals for 1971–2000 for Canada (highest values are in italics, and lowest values are in boldface).

	ME			MAE			RMSE		
	D20	D05	D05adj	D20	D05	D05adj	D20	D05	D05adj
Max temperature (°C)									
Winter	0.23	0.29	<i>0.32</i>	0.49	<i>0.53</i>	0.43	0.84	<i>0.94</i>	0.68
Spring	0.06	0.11	<i>0.24</i>	<i>0.48</i>	0.40	0.40	<i>0.65</i>	0.52	0.52
Summer	0.09	0.07	<i>0.11</i>	<i>0.47</i>	0.37	0.33	<i>0.72</i>	0.52	0.49
Autumn	0.07	0.13	<i>0.17</i>	<i>0.31</i>	0.27	0.27	<i>0.47</i>	0.37	0.43
All	0.11	0.15	<i>0.21</i>	<i>0.44</i>	0.39	0.36	<i>0.68</i>	0.62	0.54
Min temperature (°C)									
Winter	<i>0.26</i>	0.20	0.24	<i>0.93</i>	0.87	0.77	<i>1.29</i>	1.22	1.04
Spring	<i>−0.07</i>	−0.06	0.03	<i>0.60</i>	0.59	0.56	0.78	<i>0.79</i>	<i>0.79</i>
Summer	−0.15	−0.17	<i>−0.20</i>	0.60	<i>0.62</i>	<i>0.62</i>	<i>0.80</i>	0.78	0.79
Autumn	0.09	0.13	<i>0.10</i>	<i>0.71</i>	0.69	0.67	<i>0.90</i>	0.85	0.87
All	0.03	0.03	<i>0.05</i>	<i>0.71</i>	0.69	0.66	<i>0.97</i>	0.93	0.88
Precipitation (mm)									
Winter	2.2	1.0	−0.6	7.5	6.6	5.8	<i>10.5</i>	9.0	8.1
Spring	−0.3	−0.9	<i>−2.2</i>	<i>4.8</i>	4.5	4.5	7.8	6.6	7.1
Summer	0.8	0.5	<i>−2.2</i>	<i>5.0</i>	4.7	4.4	<i>6.8</i>	6.0	5.9
Autumn	1.2	0.1	<i>−1.7</i>	<i>5.3</i>	4.8	4.3	<i>7.6</i>	6.6	6.3
All	1.0	0.2	<i>−1.7</i>	<i>5.7</i>	5.2	4.7	<i>8.3</i>	7.1	6.9

Nevertheless, it is possible that the linear regressions have errors in the estimated normals that are better removed by the data smoothing applied by SPLINB to each dataset. Indeed, Hutchinson and Bischof (1983) and Hutchinson (1995) have found that spatial spline smoothing can be very effective in reducing errors in short-period means. The best way to determine this is to assess the predictive errors of the fitted surfaces against the long-period normals of the spatially representative withheld data. This test is independent of the errors contained in each of the three datasets.

b. Withheld-station tests

SPLINB was applied to the same data as above, but with 48 spatially representative stations, as shown in Fig. 2, withheld from each dataset. The withheld stations have very complete records over the period 1971–2000 so that their normals could be expected to have close to zero error. The majority of these stations are principal stations, and 14 are ordinary volunteer climate stations. The monthly predictive error statistics on the basis of the withheld data are summarized by season and year in Table 3. This table reports the bias or mean error (ME), the mean absolute error (MAE), and the root-mean-square error (RMSE). The MAE and the RMSE are the key measures of predictive accuracy for each dataset. It is noted that the RtGCVs and RtMSRs for the withheld data analyses (not shown) are very similar to the values

presented in Table 2, agreeing to within 1% or 2%. This confirms that the omitted 48 stations did not markedly reduce the overall predictive accuracies of the fitted surfaces.

1) MAXIMUM TEMPERATURE

As shown in Table 3, there is a small positive bias, or ME, for the withheld stations in all months. This ranges from around 0.1°C in summer to about 0.3°C in winter. D05adj gives rise to the largest bias in all seasons.

The MAE and RMSE values show very similar results. For most seasons, the adjusted dataset D05adj yields the best MAE and RMSE values, whereas the D05 dataset yields the best MAE and RMSE values in spring and autumn, confirming that for some seasons spatial smoothing is very competitive with the individual station linear regressions in adjusting short-period means. The smallest dataset with the best long-term stations (D20) yields the largest MAE and RMSE values in most seasons. Overall, the adjusted dataset D05adj yields the smallest predictive errors, with MAE ranging from 0.27°C in the autumn to 0.43°C in winter. The annual-summary MAE for the D05adj data is 8% smaller than the annual-summary MAE for the D05 data and is 18% smaller than the annual-summary MAE for the D20 data.

The largest differences at individual withheld stations are for stations in northern British Columbia and the

Yukon in the winter months and at coastal locations in the summer months. In the case of northern British Columbia and the Yukon, this result reflects the relatively sparse data coverage in this part of the country. This sparseness is compounded by the complex terrain and a sharp transition from coastal climate along the Pacific Ocean to the continental climate of Fort Nelson in extreme northeastern British Columbia. With respect to the coastal stations in the summer, cold ocean waters give rise to a sea-breeze circulation that reduces the maximum temperatures relative to inland stations. Thus, for the D20 data, Yarmouth on the east coast has larger positive residuals, because the estimated value there is strongly influenced by nearby inland stations. On the other hand, Sydney, Nova Scotia, has negative residuals, reflecting the impact of a different spatial configuration of nearby cold coastal stations.

2) MINIMUM TEMPERATURE

Table 3 shows for all three datasets a positive bias or ME of around 0.2°C for the winter months and an almost 0.2°C negative bias in the summer months.

As in the maximum temperature analyses, the MAE and the RMSE show similar behavior across all three datasets, and there is somewhat similar performance between the D05 and D05adj datasets, with the D05adj dataset again yielding the overall best results. In particular, the MAE and RMSE for the D05adj dataset tend to be lowest in the winter. There is little difference in performance among all three datasets in the summer and in the spring. In summary, the MAEs for the D05adj dataset range from 0.56°C in spring to 0.77°C in winter, systematically larger than the corresponding MAEs for maximum temperature. The annual-summary MAE for the D05adj data is 4% smaller than the annual-summary MAE for the D05 data and is 7% smaller than the annual-summary MAE for the D20 data.

A detailed examination of the coastal stations shows no substantive influence of the nearby cold water bodies on the spatial estimation of minimum temperature in coastal areas in summer, unlike for the maximum temperature. At night, the sea-breeze circulation reverses or is nonexistent so that minimum temperatures even in coastal areas usually reflect a land environment. Minimum temperature is affected more by local effects such as cold-air drainage into frost hollows, and so it is expected that the spatial predictability of minimum temperature would be more of a challenge than that of maximum temperature, which is solar driven.

3) PRECIPITATION

Table 3 shows small positive biases for the D20 and D05 datasets of 0.1–2 mm through most of the year,

whereas the D05adj dataset gives rise to a negative bias in all seasons. Inspection of individual withheld stations reveals that most of this negative bias is attributable to stations along the north shore of the Gulf of St. Lawrence and to all four stations from Newfoundland—most noticeably, St. John's A.

Table 3 shows that, as for maximum and minimum temperature, the adjusted precipitation dataset D05adj gives rise to the smallest MAE and RMSE values overall and the dataset D20 gives rise to the largest values. As with maximum temperature, the D05 data are competitive with the D05adj data in spring, particularly in terms of the RMSE. The overall improvement in MAE and RMSE between the D05 and D05adj datasets is therefore modest, but both datasets consistently outperform the D20 dataset. In summary, the MAE for the D05adj dataset ranges from 4.3 mm in the autumn to 5.8 mm in winter. The annual-summary MAE for the D05adj data is around 10% smaller than the annual-summary MAE for the D05 data and is 18% smaller than the annual-summary MAE for the D20 data.

The larger withheld residuals in winter reflect the difficulties in measuring precipitation in winter as well as the sparse data network and complex terrain on the west coast. The fitted models underestimated the data values for Vancouver Island by as much as 100 mm in mid-winter. The noticeably larger errors on the west coast suggest that there are too few stations to adequately represent the precipitation climate of Canada there.

4. Discussion

Canada is a large country with a great variety of topographic and regional influences on climate. Much of northern Canada has a sparse network of climate stations, and only along Canada's southern extremity does the network approach WMO recommended standards for station density. Even here there are notable exceptions such as at high elevations over the complex terrain of British Columbia and over northwestern Ontario north of Lake Superior. A full 30 years of data at all existing stations from 1971 to 2000 would be the ideal, but less than 20% of the climate stations in Canada meet the WMO requirement for a minimum of three missing successive years or any five years during this normals period. Despite these limitations, Canada-wide (gridded) robust and spatially reliable models of monthly normal climate are highly desirable and of great interest.

This study assessed the ability of two datasets, D05 and D05adj, to improve on the predictive accuracy of climate normals interpolated from the substantially smaller, but more accurate, D20 dataset that is based on stations with at least 20 years of data. The D05 and

D05adj datasets were based on stations with at least 5 years of data, with the D05adj data adjusted by linear regression with nearby long-term stations. The ANUSPLIN analyses of each full dataset confirmed that the D05 dataset, consisting of the unadjusted data, was the noisiest dataset. This was indicated by the systematically larger RtMSRs of the fitted splines. The similar residuals for the D20 and D05adj datasets also suggest that the means determined from records extended by linear regression had accuracy that is similar to that of the means in the D20 dataset.

Important is that assessing the accuracy of the surfaces fitted to the three datasets with GCV is not straightforward, since the GCV depends on both the accuracy of the interpolated surfaces and the errors in the data themselves. It is also subject to some bias resulting from the uneven spacing of the Canadian data network. If one accepts the similar errors of the D20 and D05adj data, the generally smaller RtGCVs for the D05adj dataset indicate that the larger D05adj dataset has given rise to more accurate interpolated climate normals than have the D20 data. The larger signals for the D05adj analyses also indicate that more information has been extracted from these data. The smaller RtGCV for maximum temperature for the D20 dataset in the autumn appears to be due to the smaller residuals from the D20 data in this season. This is confirmed by the withheld-data tests, which indicate the clear superiority of the D05adj dataset over the D20 dataset.

It remains then to assess the relative merits of the D05 and D05adj datasets in supporting interpolated climate normals. Overall, the withheld-data tests indicate that the D05adj was superior to the D05 dataset, but this was not the case for all seasons, and there were somewhat different results for the different climate variables for particular seasons. For maximum temperature, the D05adj data were generally superior to the D05 data except for spring and autumn. The better performance of the D05 data in spring and autumn is consistent with the broader synoptic patterns and associated longer spatial correlation scales observed in these seasons (Milewska and Hogg 2001). The broader spatial correlation scales would assist the spatially interpolated smoothed values to overcome the errors in the short-term means, and in spring and autumn this effect appears to be very competitive with the linear regressions with nearby long-term stations in the D05adj dataset. The better performance of the D05adj data in winter may have been also assisted by the stronger positive temporal trends that have been observed in temperatures for the winter months over the period of 1971–2000 (Zhang et al. 2000).

The comparisons between the D05 and D05adj minimum temperature and precipitation data are broadly

similar to those for maximum temperature but are probably somewhat masked by greater observational difficulties for these variables. For minimum temperature, the D05adj data are generally superior in winter but display mixed results in the other three seasons, and in fact no difference is observed among all three datasets in summer. For precipitation, D05adj is also generally superior in winter, again perhaps reflecting stronger temporal trends in this season (Zhang et al. 2000). As was seen for maximum temperature, D05 is also competitive with D05adj in spring, in terms of both MAE and RMSE, but with the D05adj dataset being superior overall.

5. Conclusions

In summary, the withheld-data tests indicate that the two larger datasets D05 and D05adj are generally superior to the D20 dataset in supporting spatial interpolation of 30-yr climate normals. The D20 dataset has the poorest ability to represent the full range of climate conditions over Canada. The regression-adjusted D05adj dataset produced the best overall results, particularly for maximum temperature and precipitation, with summary predictive errors for the interpolated normals being 4% and 10% smaller than the summary predictive errors for the unadjusted D05 dataset and 7% and 18% smaller than the summary predictive errors for the D20 dataset. Also for maximum temperature and precipitation, however, the unadjusted D05 data were very competitive with the D05adj data in spring and autumn, when longer spatial correlation scales are apparent. The daily spatial analyses by Hutchinson et al. (2009) similarly found smaller predictive errors for precipitation and maximum temperature in the autumn. The larger temporal trends in winter in climate data over the last half century are likely to have also contributed to the markedly superior performance of the adjusted D05adj dataset across all climate variables in winter.

Thus, for the unevenly distributed Canadian data network across a broad range of climate conditions, more data are better, provided care has been taken to minimize the effects of unrepresentative short records. The models represent the three climatological elements well in most parts of the country, but prediction errors are relatively larger in particularly data-sparse areas with complex climate patterns associated with complicated topography and proximity to coastlines. For example, the sparse network in northern Canada may not be able to capture some features and gradients, and thus errors will generally be larger in those areas. Elsewhere, the ANUSPLIN estimates of normals for all three

climate elements at withheld stations are relatively good, as evidenced by the summary predictive errors in Table 3 and in comparison with summary predictive errors documented by other studies (Price et al. 2000; Daly et al. 2008).

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